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In [2]: from pathlib import Path
import numpy as np
import pandas as pd

from sklearn.linear_model import Lars
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import LeaveOneOut, cross_val_predict
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

# Load enzyme data
csv_path = Path("Wold_DDGTs.csv")
if not csv_path.exists():
    csv_path = Path(r"C:\Users\Mikey\Desktop\CHE891_Classwork\Process_Design\Class_

df = pd.read_csv(csv_path)
print("Data shape:", df.shape)
print(df.head())

# Predict DDGTs from physicochemical descriptors
feature_cols = ["PIE", "PIF", "DGR", "SAC", "MR", "Lam", "Vol"]
target_col = "DDGTs"

X = df[feature_cols].values
y = df[target_col].values

# Small dataset: tune sparsity using Leave-One-Out CV
loo = LeaveOneOut()
max_vars = X.shape[1]
cv_r2 = []

for n_vars in range(1, max_vars + 1):
    model_cv = Pipeline([
        ("scaler", StandardScaler()),
        ("lars", Lars(n_nonzero_coefs=n_vars))
    ])
    y_hat_loo = cross_val_predict(model_cv, X, y, cv=loo)
    cv_r2.append(r2_score(y, y_hat_loo))

best_n_vars = int(np.argmax(cv_r2)) + 1
print("\nLOOCV R^2 by # active variables:")
for n_vars, score in enumerate(cv_r2, start=1):
    print(f" {n_vars:>2d} vars: {score:+.4f}")
print(f"\nBest sparsity: {best_n_vars} active variables")

# Fit final LARS model on full data with selected sparsity
model = Pipeline([
    ("scaler", StandardScaler()),
    ("lars", Lars(n_nonzero_coefs=best_n_vars))
])
model.fit(X, y)
y_pred = model.predict(X)

r2_train = r2_score(y, y_pred)

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rmse_train = mean_squared_error(y, y_pred) ** 0.5
print(f"\nFinal model training R^2: {r2_train:.4f}")
print(f"Final model training RMSE: {rmse_train:.4f}")

# Report selected coefficients in original feature order
coef = model.named_steps["lars"].coef_
coef_table = pd.DataFrame({
    "feature": feature_cols,
    "coefficient": coef,
    "selected": np.abs(coef) > 1e-12
})
print("\nLARS coefficients:")
print(coef_table)

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Data shape: (19, 8)

	PIE	PIF	DGR	SAC	MR	Lam	Vol	DDGTS
0	0.23	0.31	-0.55	254.2	2.126	-0.02	82.2	8.5
1	-0.48	-0.60	0.51	303.6	2.994	-1.24	112.3	8.2
2	-0.61	-0.77	1.20	287.9	2.994	-1.08	103.7	8.5
3	0.45	1.54	-1.40	282.9	2.933	-0.11	99.1	11.0
4	-0.11	-0.22	0.29	335.0	3.458	-1.19	127.5	6.3

LOOCV R^2 by # active variables:

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1 vars: +0.3937
2 vars: +0.2577
3 vars: +0.2482
4 vars: +0.2318
5 vars: -0.0548
6 vars: -0.4921
7 vars: -0.5309

```

Best sparsity: 1 active variables

Final model training R^2: 0.5016

Final model training RMSE: 1.8950

LARS coefficients:

	feature	coefficient	selected
0	PIE	0.000000	False
1	PIF	1.729016	True
2	DGR	0.000000	False
3	SAC	0.000000	False
4	MR	0.000000	False
5	Lam	0.000000	False
6	Vol	0.000000	False

LARS would eliminate any coefficients equal to zero, but in this notebook I am predicting DDGTS from a set on information on each amino acid individually, so I don't understand why eliminating amino acids would improve the fit. Lars is selecting physiochemical descriptors as the variables, not amino acids.